
COLLEGE OF ENGINEERING, UIC
ARTIFICIAL INTELLIGENCE 1, CS 411

I. Instructor & Course Details

Instructor Name: Piotr Gmytrasiewicz

Email address: piotr@uic.edu

Drop-In Office Hours, virtual: Monday, 3pm, over zoom, email me for time slot

TA name: ?

Email address:

Drop-In Office Hours (in-person or virtual): Wednesday, 3:15—5:15pm?

Blackboard Course Site: Ugrad -

https://uic.blackboard.com/ultra/courses/_289914_1/cl/outline

Grad: https://uic.blackboard.com/ultra/courses/_289912_1/cl/outline

Students are expected to log into the course site regularly to learn about any developments related to the course, upload assignments, and communicate with classmates. For all technical questions about Blackboard, email the Learning Technology Solutions team at LTS@uic.edu.

Course Modality and Schedule

Mon, Wed, Fri, 10:00 – 10:50am, at LC E1, in person

Catalog Course Description and Prerequisite/corequisite Statement

Growth Mindset:

Course materials and assignments can be complex and challenging, but they are crucial to your intellectual and personal growth and development. There are times you may need extra help. Students who attend class consistently, complete all assignments, thoughtfully engage with feedback on work, develop good study strategies, visit the tutoring center, and contact faculty when they are struggling can develop a thorough understanding of the course material and ultimately succeed in the course!

Course Goals and Learning Outcomes

We will study the fundamental questions relating to intelligence, definitions of artificial intelligence, intelligent agent architectures and types of environments they operate in, search

techniques, knowledge representation using propositional and first order logic, probabilities, automated inference techniques, sequential decision-making (decision-theoretic planning), and machine learning.

Students will learn to identify, formulate and solve problems related to AI by applying principles of logic and mathematics and be able to communicate them. They will be exposed to some ethical issues posed by AI and its global context, and acquire ability to continually learn and improve their skills and understanding.

Required and Recommended Course Materials

Textbook: S. Russell and P. Norvig “Artificial Intelligence: A Modern Approach”, fourth edition, Prentice Hall.

III. COURSE POLICIES & CLASSROOM EXPECTATIONS

Grading Policy and Point Breakdown:

Homeworks: 50%

Homeworks are to be completed individually on your own.

Midterm: 15%

Final: 35%

Policy for Missed or Late Work

HWs will be usually due (almost) every Friday at class time, sharp. 48 hrs late submission -25%

Attendance / Participation Policy

Policy:

Please email me if you face an unexpected situation that may impede your attendance, participation in required class and exam sessions, or timely completion of assignments.

Other Course Policies

Academic Integrity

As a student and member of the UIC community, you are expected to adhere to the [Community Standards of academic integrity](#), accountability, and respect. Please review the [UIC Student Disciplinary Policy](#) for additional information.

Email Expectations

Students are responsible for all information instructors send to your UIC email and Blackboard accounts. Faculty messages should be regularly monitored and read in a timely fashion.

IV. COURSE SCHEDULE

Weekly Schedule of Class Topics (Outline)

Week 1: Intro and definition of AI, Chapt 1
Week 2: Agents and Environments, Chapt 2
Week 3: Agents and Environments, Chapt 2
Week 4: Uninformed Search, Chapt 3
Week 5: Informed Search, Chapt 3
Week 6: Search in Complex Environments, Chapt 4
Week 7: Adversarial Search, Chapt 5
Week 8: Propositional Logic, Chapt 7
Week 9: First Order Logic, Chapt 8
Week 10: Inference in First Order Logic, Chapt 9
Week 11: Probability, Chapt 12
Week 12: Bayesian Reasoning, Chapt 13
Week 13: Sequential Decisions, Chapt 17
Week 14: Learning of Decision Trees, Chapt 19
Week 15: Deep Learning, LLMs, Chapt 21
Week 16: Final exam week

Disclaimer

This syllabus is intended to give the student guidance on what may be covered during the semester and will be followed as closely as possible. However, as the instructor, I reserve the right to modify, supplement, and make changes as course needs arise. I will communicate such changes in advance through in-class announcements and in writing via Blackboard Announcements.

V. ACCOMMODATIONS

Disability Accommodation Procedures

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc.uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.

Religious Accommodations

Following [campus policy](#), if you wish to observe religious holidays, you must notify me by the tenth day of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify me at least five days before you will be absent. Please submit [this form](#) by email with the subject heading: **"YOUR NAME: Requesting Religious Accommodation."**

VI. CLASSROOM ENVIRONMENT

Inclusive Community

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If aspects of this course result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify me as soon as possible.

Name and Pronoun Use

If your name does not match the name on my class roster, please let me know as soon as possible. My pronouns are [he/him](#). I welcome your pronouns if you would like to share them with me. For more information about pronouns, see this page: <https://www.mypronouns.org/what-and-why>.

Community Agreement/Classroom Conduct Policy

- Be present by turning off cell phones and removing yourself from other distractions.